

OPERATING SYSTEMS CONCEPT – LAB

EXP – 07: PRODUCER CONSUMER PROBLEM SOLUTION USING SEMAPHORE

Producer – Consumer Problem in C

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <semaphore.h>
4 #define BUFFER_SIZE 5
5
6 int buffer[BUFFER_SIZE];
7 int in = 0;
8 int out = 0;
9 sem_t empty, full, mutex;
10
11 void producer(void *arg) {
12     int item;
13     for (int i = 1; i <= BUFFER_SIZE; i++) {
14         item = i * 10;
15         sem_wait(&empty);
16         sem_wait(&mutex);
17         buffer[in] = item;
18         printf("Produced: %d\n", item);
19         in = (in + 1) % BUFFER_SIZE;
20         sem_post(&mutex);
21         sem_post(&full);
22     }
23     return NULL;
24 }
25
26 void consumer(void *arg) {
27     int item;
28     for (int i = 1; i <= BUFFER_SIZE; i++) {
29         sem_wait(&full);
30         sem_wait(&mutex);
31         item = buffer[out];
32         printf("Consumed: %d\n", item);
33         out = (out + 1) % BUFFER_SIZE;
34         sem_post(&mutex);
35         sem_post(&empty);
36     }
37     return NULL;
38 }
39
40 int main() {
41     pthread_t p, c;
42     sem_init(&empty, 0, BUFFER_SIZE);
43     sem_init(&full, 0, 0);
44     sem_init(&mutex, 0, 1);
45     pthread_create(&p, NULL, producer, NULL);
46     pthread_create(&c, NULL, consumer, NULL);
47     pthread_join(p, NULL);
48     pthread_join(c, NULL);
49     sem_destroy(&empty);
50     sem_destroy(&full);
51     sem_destroy(&mutex);
52     return 0;
53 }
```

snordeen@localhost:~/os-lab/producer_consumer — vim pc.c

Activities Terminal Apr 20 14:15

snordeen@localhost:~/os-lab/producer_consumer\$ gcc pc.c -o pc -lpthread

snordeen@localhost:~/os-lab/producer_consumer\$./pc

```
Produced: 10
Produced: 20
Consumed: 10
Produced: 30
Consumed: 20
Produced: 40
Consumed: 30
Produced: 50
Consumed: 40
Consumed: 40
Consumed: 50
snordeen@localhost:~/os-lab/producer_consumer$
```